

```
status=1
while [ $status -ne 0 ]
do
    lynx -cmd_script=dir_in <URL of login page>
    status=$?      # $? returns 0 if previous command was executed
done
successfully,
done
```

That makes our final **login.sh** script file ready for use.

Downloading

Now that the system is connected, the next step is downloading the required files. We will use **wget** as our download utility, which accepts the URL of the file to be downloaded as its argument. We can invoke **wget** as follows:

```
wget <URL>
```

You can specify the folder to which **wget** saves the downloaded material. Simply add an option, **--directory-prefix=<target folder>**. For example,

```
wget --directory-prefix=/media/new_volume <URL>
```

Here, **/media/new_volume** is the mount point for my backup drive, which contains all my downloaded files.

However, before we create the download scheduler script, we should create a collection of URLs of the files we wish to download, one after the other.

Let's now write another script called **download.sh**, which contains a set of **wget** commands pointing to the required URLs. Here's an example:

```
#download.sh
#!/bin/bash
wget --directory-prefix=<target folder> <URL of file 1>
wget --directory-prefix=<target folder> <URL of file 2>
##(append as many wget commands as you wish)
wget --directory-prefix=<target folder> <URL of file n>
```

The above script is, naturally, the most important part of our **start.sh** script.

It includes **login.sh** and **download.sh** and, of course, is the script you need to run in any one of the three terminal windows as mentioned earlier. The function of **start.sh** is simple—run **login.sh** and then **download.sh**. Here's what it should look like:

```
#start.sh
#!/bin/bash
bash login.sh
bash download.sh
```

Logging out

After downloading the necessary files, the next job is to log out. So we are on our way to write the **logout.sh**

script. The steps to log out are similar to logging in. We will create a file called **dir_out** to store the keystrokes we need in order to log out.

As with logging in, the procedure to log out may vary with ISPs and connection-plans. For instance, in my case, I need to re-submit my user name and password in order to log out. My ISP doesn't have an independent logout page. So when I open my browser with the URL of my ISP, even when I am online, I'll have to enter the user credentials to access the logout page. However, for others, this might not be the case. If your ISP's servers can auto-detect the connection status and pull up the logout page instead, when you invoke the browser with the ISP's URL, it's easy to create **dir_out**, as you only need to run the following:

```
lynx -cmd_log=dir_out <URL>
```

...and keep hitting the **Tab** key till you reach the 'logout' link; finally, press **Enter** to log out.

Those in a similar situation can run the same command as above, resubmit a login ID after filling the text fields, and submit the logout after highlighting the logout link by moving over preceding links in the page. Now quit from the browser and provide the confirmation. Our **dir_out** file has now been created.

Again, it is possible that the logout isn't successful at the first attempt. So let's bring in the magic of the **while** loop again and our final **logout.sh** script should look like what follows:

```
#logout.sh
#!/bin/bash
value=1
while [ $value -ne 0 ]
do
    lynx -cmd_script=dir_out <URL of ISP>
    value=$?
done
```

Shutting down the system

After essential downloads and a successful logout, it would be wise to shutdown the system—we should save electricity, seriously. Here, we can make use of the **halt** command. Note that you can't be careless while you invoke **halt**. It will initiate a forced shutdown and all your unsaved data will be lost. Also, it can only be executed by the root user. Save the command in a separate file called **shutdown.sh** with the following data:

```
#shutdown.sh
#!/bin/bash
halt
```

That's it; all the essential script files are now ready. It's time to put these in a schedule.